

## 1 Summary

Model (Run-name): model3 (test)

Iteration: 13

**TAC: 63974.04503894617**

Took: 721.3726115226746s

Total Time: 2584.5096213817596s

Epsilons:

- $\varepsilon_{LMTD}$ : 1e-06
- $\varepsilon_{Balancing}$ : 1e-06
- $\varepsilon_{Area}$ : 1e-06
- $\varepsilon_{Beta}$ : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: [http://www.gamsworld.org/minlp/minlplib2/html/heatexch\\_gen3.html](http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen3.html)

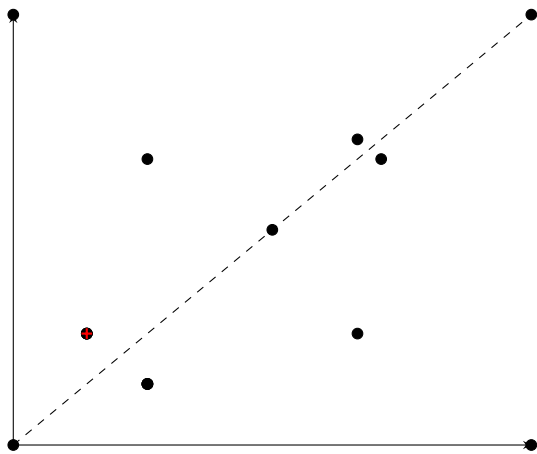
## 2 Results

| Variable          | Value              | Variable          | Value                | Variable        | Value              |
|-------------------|--------------------|-------------------|----------------------|-----------------|--------------------|
| $z_{1,1,2}$       | 1.0                | $dt_{3,1,1}$      | 66.7                 | $q_{1,1,2}$     | 586.96             |
| $z_{2,2,2}$       | 1.0                | $dt_{3,1,2}$      | 77.77652916073785    | $q_{2,2,2}$     | 647.2099999999999  |
| $z_{2,5,2}$       | 1.0                | $dt_{3,5,2}$      | 10.60810810810895    | $q_{2,5,2}$     | 530.45             |
| $z_{3,1,1}$       | 1.0                | $dt_{3,5,3}$      | 53.20540540540817    | $q_{3,1,1}$     | 173.03999999998723 |
| $z_{3,5,2}$       | 1.0                | $dt_{4,3,2}$      | 50.0                 | $q_{3,5,2}$     | 1013.8400000000134 |
| $z_{4,3,2}$       | 1.0                | $dt_{4,3,3}$      | 111.1                | $q_{4,3,2}$     | 1539.72            |
| $z_{5,4,2}$       | 1.0                | $dt_{5,4,2}$      | 22.2                 | $q_{5,4,2}$     | 1634.85            |
| $z_{cu,3}$        | 1.0                | $dt_{5,4,3}$      | 24.33559322033713    | $q_{cu,3}$      | 1197.4             |
| $z_{cu,5}$        | 1.0                | $dt_{cu,3}$       | 106.50540540540817   | $q_{cu,5}$      | 724.5599999999672  |
| $A_{1,1,2}^\beta$ | 6.886621175724489  | $dt_{cu,5}$       | 66.53559322033713    | $t_{1,1}^{(H)}$ | 160.0              |
| $A_{2,2,2}^\beta$ | 7.746030103163966  | $f_{1,1,2}^H$     | 8.8                  | $t_{1,2}^{(H)}$ | 160.0              |
| $A_{2,5,2}^\beta$ | 3.5642637680418505 | $f_{2,2,2}^H$     | 5.825472547254702    | $t_{1,3}^{(H)}$ | 93.3               |
| $A_{3,1,1}^\beta$ | 0.4281862564459283 | $f_{2,5,2}^H$     | 4.774527452745158    | $t_{2,1}^{(H)}$ | 248.9              |
| $A_{3,5,2}^\beta$ | 9.166515043502212  | $f_{3,1,1}^H$     | 1.0741154562382824   | $t_{2,2}^{(H)}$ | 248.9              |
| $A_{4,3,2}^\beta$ | 6.6757786012311815 | $f_{3,5,2}^H$     | 7.646039652272265    | $t_{2,3}^{(H)}$ | 137.8              |
| $A_{5,4,2}^\beta$ | 14.143774763563345 | $f_{4,3,2}^H$     | 12.6                 | $t_{3,1}^{(H)}$ | 226.7              |
| $A_{cu,3}^\beta$  | 6.0263880678109905 | $f_{5,4,2}^H$     | 12.264441110277568   | $t_{3,2}^{(H)}$ | 215.00810810810896 |
| $A_{cu,5}^\beta$  | 5.324394542067401  | $f_{1,1,2}^C$     | 5.869600000000002    | $t_{3,3}^{(H)}$ | 146.50540540540817 |
| $A_{1,1,2}$       | 24.927400386350968 | $f_{2,2,2}^C$     | 6.1                  | $t_{4,1}^{(H)}$ | 271.1              |
| $A_{2,2,2}$       | 30.93278282491083  | $f_{2,5,2}^C$     | 4.774527452745016    | $t_{4,2}^{(H)}$ | 271.1              |
| $A_{2,5,2}$       | 14.023793787177482 | $f_{3,1,1}^C$     | 7.6                  | $t_{4,3}^{(H)}$ | 148.9              |
| $A_{3,1,1}$       | 1.221214580613199  | $f_{3,5,2}^C$     | 9.125472547254844    | $t_{5,1}^{(H)}$ | 198.9              |
| $A_{3,5,2}$       | 41.09845581737977  | $f_{4,3,2}^C$     | 8.4                  | $t_{5,2}^{(H)}$ | 198.9              |
| $A_{4,3,2}$       | 23.668654733871367 | $f_{5,4,2}^C$     | 17.3                 | $t_{5,3}^{(H)}$ | 106.53559322033713 |
| $A_{5,4,2}$       | 82.71968770371991  | $RecLMTD_{1,1,2}$ | 0.03609835812479837  | $t_{1,1}^{(C)}$ | 160.0              |
| $A_{cu,3}$        | 20.61398815516275  | $RecLMTD_{2,2,2}$ | 0.04062493688474253  | $t_{1,2}^{(C)}$ | 137.23157894736954 |
| $A_{cu,5}$        | 16.23492858720698  | $RecLMTD_{2,5,2}$ | 0.0224719101123592   | $t_{1,3}^{(C)}$ | 60.0               |
| $dt_{1,1,2}$      | 22.76842105263163  | $RecLMTD_{3,1,1}$ | 0.021362771168410573 | $t_{2,1}^{(C)}$ | 221.7              |
| $dt_{1,1,3}$      | 33.3               | $RecLMTD_{3,5,2}$ | 0.034456805259975104 | $t_{2,2}^{(C)}$ | 221.7              |
| $dt_{2,2,2}$      | 27.2               | $RecLMTD_{4,3,2}$ | 0.01306624355323738  | $t_{2,3}^{(C)}$ | 115.6              |
| $dt_{2,2,3}$      | 22.2               | $RecLMTD_{5,4,2}$ | 0.04300806468370917  | $t_{3,1}^{(C)}$ | 221.1              |
| $dt_{2,5,2}$      | 44.5               | $RecLMTD_{cu,3}$  | 0.014633280384072729 | $t_{3,2}^{(C)}$ | 221.1              |
| $dt_{2,5,3}$      | 44.5               | $RecLMTD_{cu,5}$  | 0.019045612922499823 | $t_{3,3}^{(C)}$ | 37.8               |

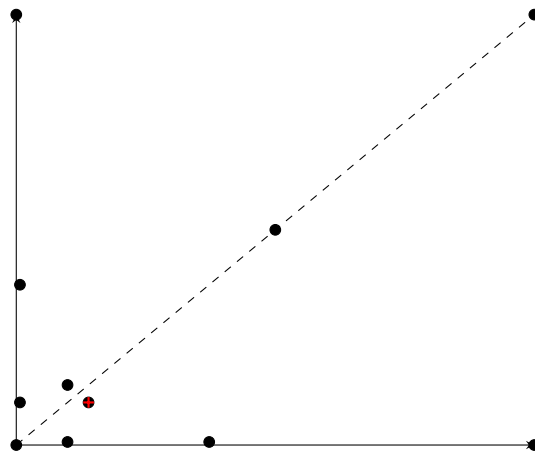
| Variable            | Value              | Variable            | Value              |
|---------------------|--------------------|---------------------|--------------------|
| $t_{4,1}^{(C)}$     | 176.7              | $b_{5,4,2}^{H,out}$ | 804.5473368342083  |
| $t_{4,2}^{(C)}$     | 176.7              | $b_{1,1,2}^{C,in}$  | 352.1760000000001  |
| $t_{4,3}^{(C)}$     | 82.2               | $b_{2,2,2}^{C,in}$  | 705.16             |
| $t_{5,1}^{(C)}$     | 204.4              | $b_{2,5,2}^{C,in}$  | 445.46341134111003 |
| $t_{5,2}^{(C)}$     | 204.4              | $b_{3,1,1}^{C,in}$  | 1042.9600000000125 |
| $t_{5,3}^{(C)}$     | 93.3               | $b_{3,5,2}^{C,in}$  | 851.4065886588769  |
| $t_{1,1,2}^H$       | 93.3               | $b_{4,3,2}^{C,in}$  | 317.52             |
| $t_{2,2,2}^H$       | 137.8              | $b_{5,4,2}^{C,in}$  | 1422.0600000000002 |
| $t_{2,5,2}^H$       | 137.8              | $b_{1,1,2}^{C,out}$ | 939.1360000000002  |
| $t_{3,1,1}^H$       | 105.6512608349287  | $b_{2,2,2}^{C,out}$ | 1352.37            |
| $t_{3,5,2}^H$       | 80.32547216088162  | $b_{2,5,2}^{C,out}$ | 975.9134113411101  |
| $t_{4,3,2}^H$       | 148.9              | $b_{3,1,1}^{C,out}$ | 1216.0             |
| $t_{5,4,2}^H$       | 74.9354156398756   | $b_{3,5,2}^{C,out}$ | 1865.2465886588902 |
| $t_{1,1,2}^C$       | 160.0              | $b_{4,3,2}^{C,out}$ | 1857.24            |
| $t_{2,2,2}^C$       | 221.7              | $b_{5,4,2}^{C,out}$ | 3056.91            |
| $t_{2,5,2}^C$       | 167.59785460886502 |                     |                    |
| $t_{3,1,1}^C$       | 160.0              |                     |                    |
| $t_{3,5,2}^C$       | 191.29173393039315 |                     |                    |
| $t_{4,3,2}^C$       | 221.1              |                     |                    |
| $t_{5,4,2}^C$       | 176.7              |                     |                    |
| $b_{1,1,2}^{H,in}$  | 1408.0             |                     |                    |
| $b_{2,2,2}^{H,in}$  | 1449.960117011698  |                     |                    |
| $b_{2,5,2}^{H,in}$  | 1188.3798829882699 |                     |                    |
| $b_{3,1,1}^{H,in}$  | 243.5019739292186  |                     |                    |
| $b_{3,5,2}^{H,in}$  | 1733.3571891701222 |                     |                    |
| $b_{4,3,2}^{H,in}$  | 3415.86            |                     |                    |
| $b_{5,4,2}^{H,in}$  | 2439.3973368342085 |                     |                    |
| $b_{1,1,2}^{H,out}$ | 821.0400000000002  |                     |                    |
| $b_{2,2,2}^{H,out}$ | 802.7501170116981  |                     |                    |
| $b_{2,5,2}^{H,out}$ | 657.9298829882828  |                     |                    |
| $b_{3,1,1}^{H,out}$ | 70.46197392923136  |                     |                    |
| $b_{3,5,2}^{H,out}$ | 719.5171891701086  |                     |                    |
| $b_{4,3,2}^{H,out}$ | 1876.14            |                     |                    |

### 3 *RecLMTD* Tangent Points

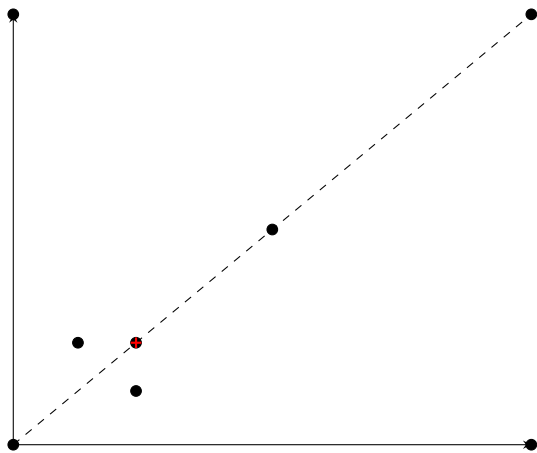
- (1, 1, 2)
  - Value: 0.036098
  - Error: 0.000000964164276
  - Relative error: 0.000026708653087
- (2, 2, 2)
  - Value: 0.040625
  - Error: 0.0000000000000001
  - Relative error: 0.0000000000000025
- (2, 5, 2)
  - Value: 0.022472
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000015
- (3, 1, 1)
  - Value: 0.021363
  - Error: 0.007492474004054
  - Relative error: -0.540181217119748
- (3, 5, 2)
  - Value: 0.034457
  - Error: 0.003398680282176
  - Relative error: 0.089780390701675
- (4, 3, 2)
  - Value: 0.013066
  - Error: 0.000000985435591
  - Relative error: 0.000075412743761
- (5, 4, 2)
  - Value: 0.043008
  - Error: 0.0000000000000002
  - Relative error: 0.0000000000000035
- $cu, 3$ 
  - Value: 0.014633
  - Error: 0.000000233609227
  - Relative error: 0.000015963987019
- $cu, 5$ 
  - Value: 0.019046
  - Error: 0.000000374191260
  - Relative error: 0.000019646724417



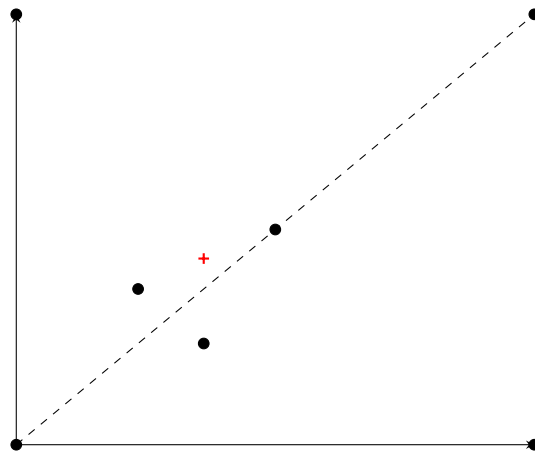
(a) RecLMTD - (1, 1, 2)



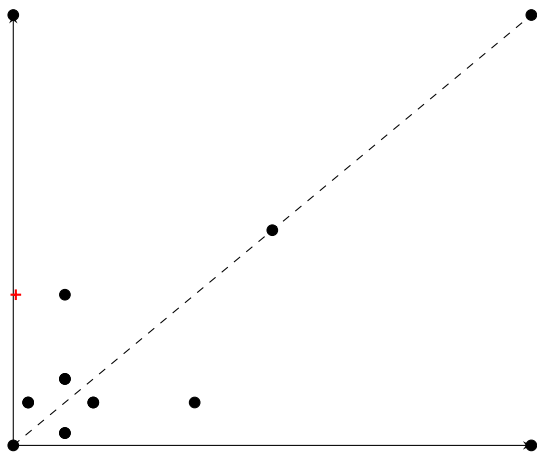
(b) RecLMTD - (2, 2, 2)



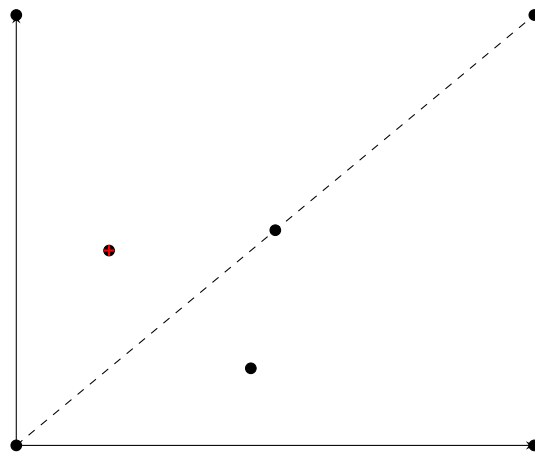
(c) RecLMTD - (2, 5, 2)



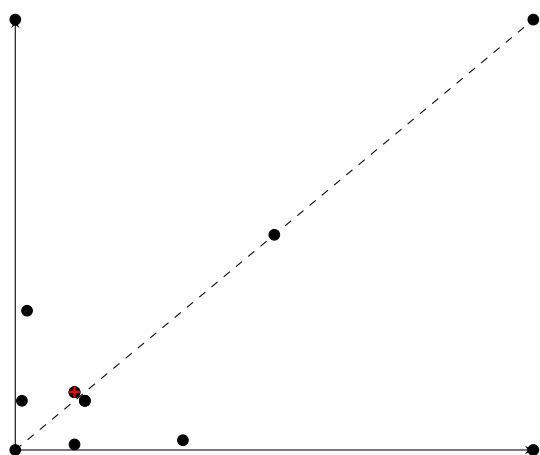
(d) RecLMTD - (3, 1, 1)



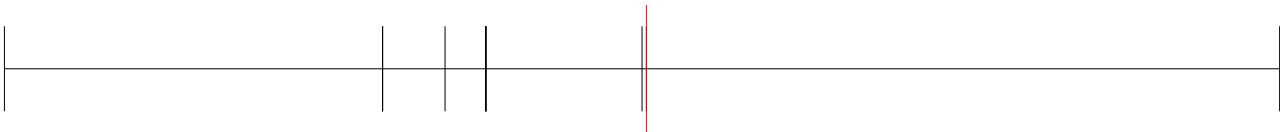
(e) RecLMTD - (3, 5, 2)



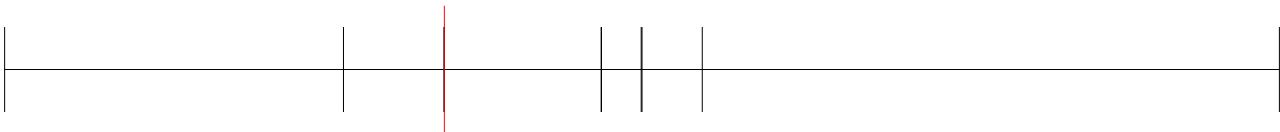
(f) RecLMTD - (4, 3, 2)



(g) *RecLMTD* - (5, 4, 2)



(a) *RecLMTD* - *cu*, 3



(b) *RecLMTD* - *cu*, 5

## 4 Balancing Breakpoints

### 4.1 $b^{H,\text{in}}$

- (1, 1, 2)
  - Value: 1408.0
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 2, 2)
  - Value: 1449.960117011698
  - Absolute error: 2.5011104298755527e-12
  - Relative error: 1.7249511904025555e-15
- (2, 5, 2)
  - Value: 1188.3798829882699
  - Absolute error: 0.0
  - Relative error: 0.0
- (3, 1, 1)
  - Value: 243.5019739292186
  - Absolute error: 2.842170943040401e-14
  - Relative error: -1.1672065310922555e-16
- (3, 5, 2)
  - Value: 1733.3571891701222
  - Absolute error: 89.39666901547935
  - Relative error: 0.05437884177843276
- (4, 3, 2)
  - Value: 3415.86
  - Absolute error: 0.0
  - Relative error: 0.0
- (5, 4, 2)
  - Value: 2439.3973368342085
  - Absolute error: 0.0
  - Relative error: 0.0

### 4.2 $b^{H,\text{out}}$

- (1, 1, 2)
  - Value: 821.0400000000002
  - Absolute error: 1.1368683772161603e-13
  - Relative error: 1.3846686851020172e-16
- (2, 2, 2)

- Value: 802.7501170116981
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 5, 2)
  - Value: 657.9298829882828
  - Absolute error: 0.0
  - Relative error: 0.0
- (3, 1, 1)
  - Value: 70.46197392923136
  - Absolute error: 43.01967830462786
  - Relative error: -0.37908928410712894
- (3, 5, 2)
  - Value: 719.5171891701086
  - Absolute error: 105.3454439405159
  - Relative error: 0.17152440625730048
- (4, 3, 2)
  - Value: 1876.14
  - Absolute error: 0.0
  - Relative error: 0.0
- (5, 4, 2)
  - Value: 804.5473368342083
  - Absolute error: 114.4936553552186
  - Relative error: -0.12457948701772369

### 4.3 $b^{C,in}$

- (1, 1, 2)
  - Value: 352.17600000000001
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 2, 2)
  - Value: 705.16
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 5, 2)
  - Value: 445.46341134111003
  - Absolute error: 0.0
  - Relative error: 0.0



- (3, 1, 1)
  - Value: 1042.9600000000125
  - Absolute error: 4.092726157978177e-12
  - Relative error: 3.924144893359423e-15
- (3, 5, 2)
  - Value: 851.4065886588769
  - Absolute error: 0.0
  - Relative error: 0.0
- (4, 3, 2)
  - Value: 317.52
  - Absolute error: 0.0
  - Relative error: 0.0
- (5, 4, 2)
  - Value: 1422.0600000000002
  - Absolute error: 0.0
  - Relative error: 0.0

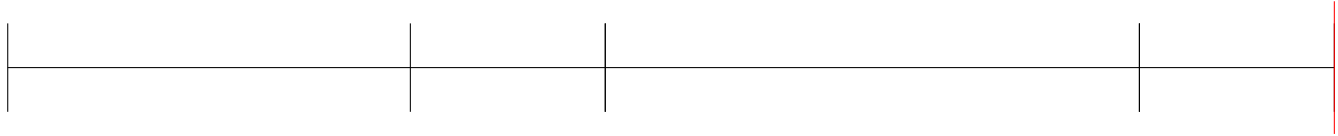
#### 4.4 $b^{C,out}$

- (1, 1, 2)
  - Value: 939.1360000000002
  - Absolute error: 1.1368683772161603e-13
  - Relative error: -1.2105471169416996e-16
- (2, 2, 2)
  - Value: 1352.37
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 5, 2)
  - Value: 975.9134113411101
  - Absolute error: 175.71285348991614
  - Relative error: 0.2195860172376834
- (3, 1, 1)
  - Value: 1216.0
  - Absolute error: 0.0
  - Relative error: 0.0
- (3, 5, 2)
  - Value: 1865.2465886588902
  - Absolute error: 119.61912216030942

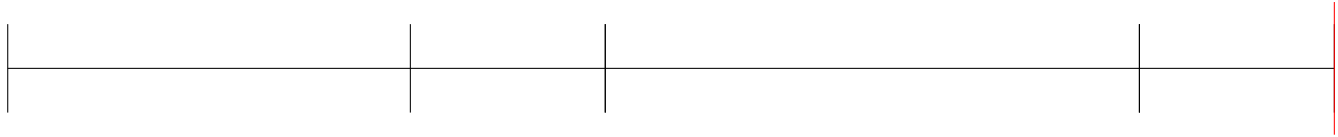
- Relative error: 0.06852500001059457
- (4, 3, 2)
  - Value: 1857.24
  - Absolute error: 0.0
  - Relative error: 0.0
- (5, 4, 2)
  - Value: 3056.91
  - Absolute error: 0.0
  - Relative error: 0.0



(a)  $t^{(H)} - (1, 2)$



(a)  $t^{(H)} - (2, 2)$



(a)  $t^{(H)} - (2, 2)$



(a)  $t^{(H)} - (3, 1)$



(a)  $t^{(H)} - (3, 2)$



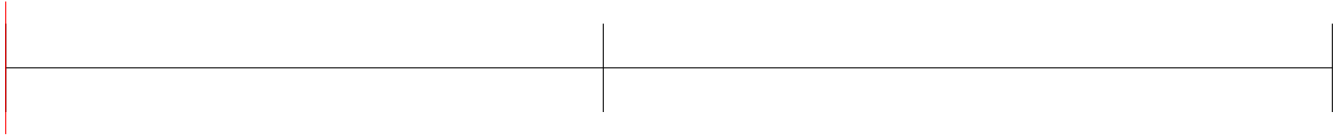
(a)  $t^{(H)} - (4, 2)$



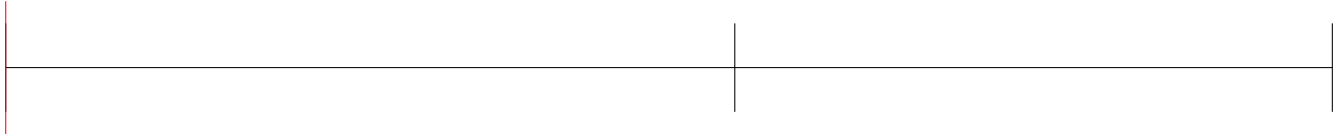
(a)  $t^{(H)} - (5, 2)$



(a)  $t^{(H)} - (1, 1, 2)$



(a)  $t^{(H)} - (2, 2, 2)$



(a)  $t^{(H)} - (2, 5, 2)$



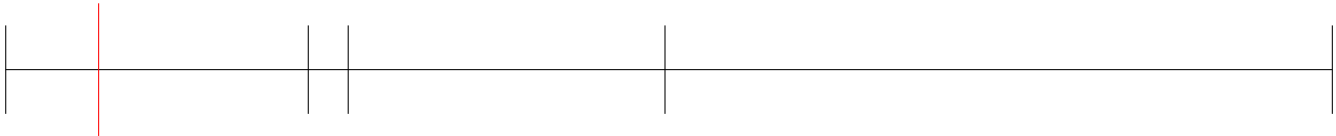
(a)  $t^{(H)} - (3, 1, 1)$



(a)  $t^{(H)} - (3, 5, 2)$



(a)  $t^{(H)} - (4, 3, 2)$



(a)  $t^{(H)} - (5, 4, 2)$



(a)  $t^{(C)} - (1, 3)$



(a)  $t^{(C)} - (2, 3)$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (1, 2)$$



$$(a) \ t^{(C)} - (5, 3)$$



$$(a) \ t^{(C)} - (3, 3)$$



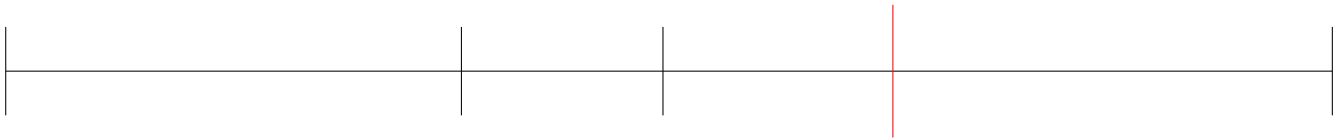
$$(a) \ t^{(C)} - (4, 3)$$



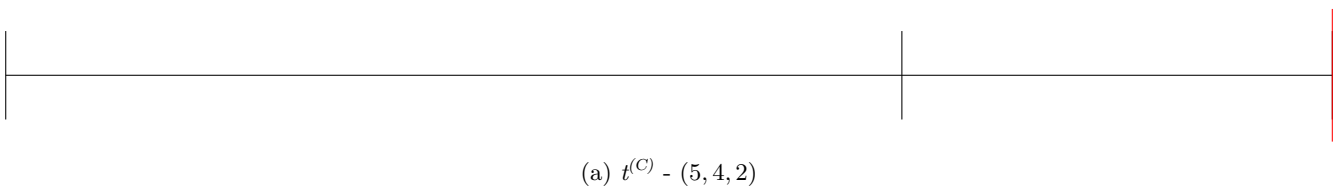
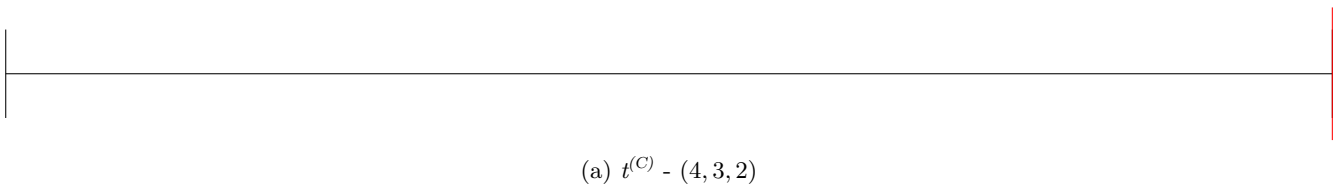
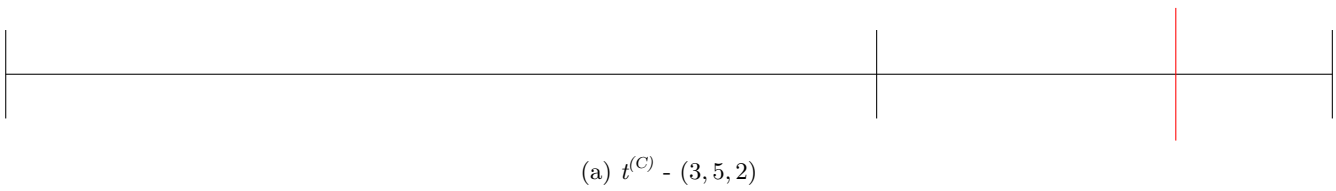
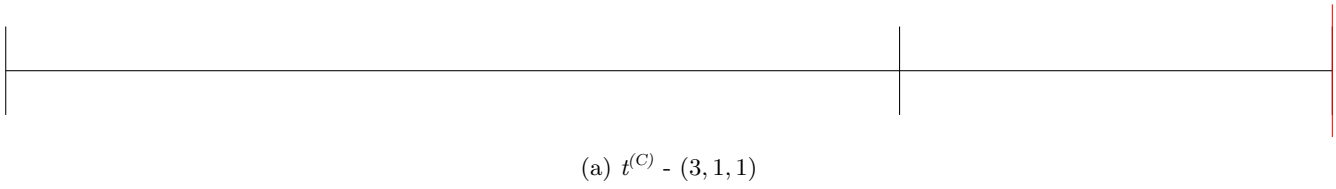
$$(a) \ t^{(C)} - (1, 1, 2)$$



$$(a) \ t^{(C)} - (2, 2, 2)$$



$$(a) \ t^{(C)} - (2, 5, 2)$$



## 5 Area Breakpoints

### 5.1 $A$

- $(1, 1, 2)$ 
  - Value: 24.927400
  - Error: 0.000002301803917
  - Relative error: 0.000000092340303
- $(2, 2, 2)$ 
  - Value: 30.932783
  - Error: 0.0000000000000007
  - Relative error: 0.0000000000000000
- $(2, 5, 2)$ 
  - Value: 14.023794
  - Error: 0.0000000000000099
  - Relative error: 0.0000000000000007
- $(3, 1, 1)$ 
  - Value: 1.221215
  - Error: 3.127742975835616
  - Relative error: 0.719193722918190
- $(3, 5, 2)$ 
  - Value: 41.098456
  - Error: 0.0000000000000966
  - Relative error: 0.0000000000000024
- $(4, 3, 2)$ 
  - Value: 23.668655
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- $(5, 4, 2)$ 
  - Value: 82.719688
  - Error: 0.0000000000000014
  - Relative error: 0.0000000000000000

### 5.2 $A_{CU}$

- $cu, 3$ 
  - Value: 20.613988
  - Error: 0.0000000000000412
  - Relative error: 0.0000000000000020
- $cu, 5$

- Value: 16.234929
- Error: 0.000000000000099
- Relative error: -0.000000000000006





(a)  $A^\beta - (1, 1, 2)$



(a)  $A^\beta - (2, 2, 2)$



(a)  $A^\beta - (2, 5, 2)$



(a)  $A^\beta - (3, 1, 1)$



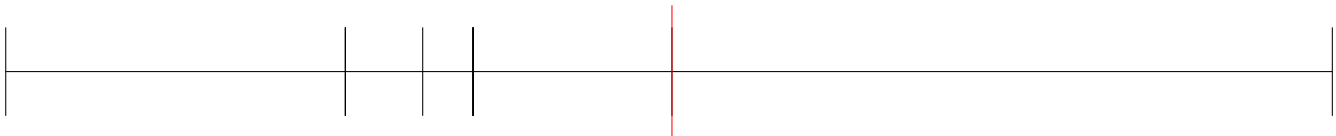
(a)  $A^\beta - (3, 5, 2)$



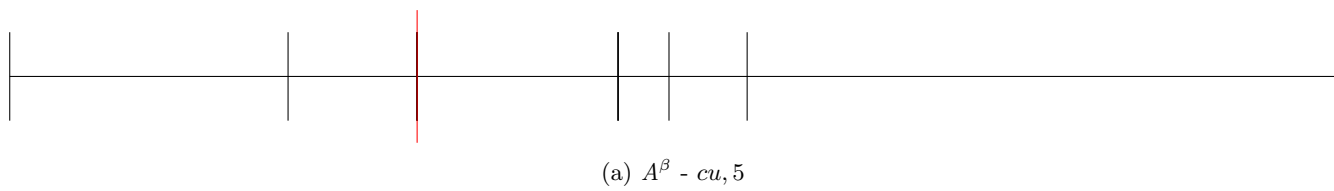
(a)  $A^\beta - (4, 3, 2)$



(a)  $A^\beta - (5, 4, 2)$



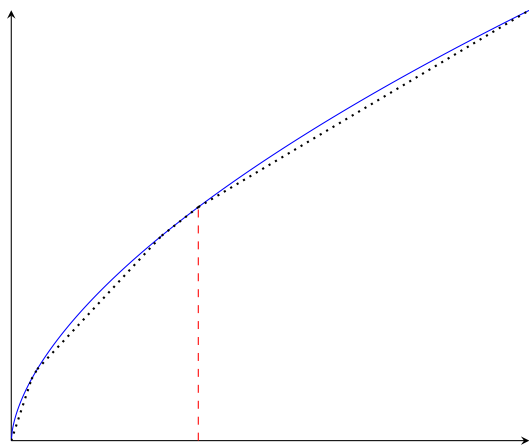
(a)  $A^\beta - cu, 3$



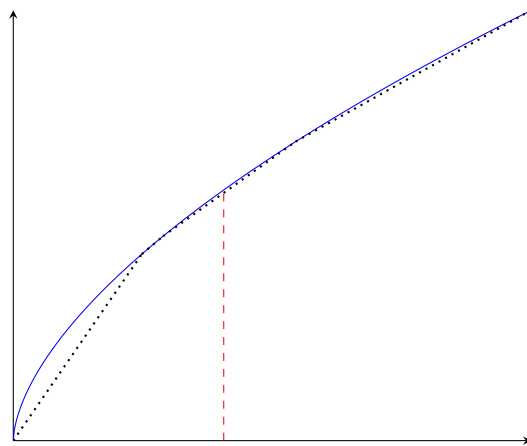
## 6 $A^\beta$ Breakpoints

- (1, 1, 2)
  - $q$ : 24.927400
  - $\hat{q}^\beta$ : 6.886621
  - $q^\beta$ : 6.886621
  - Breakpoints: [0, 2.7005882352941173, 3.460657649345137, 20.05363555217427, 24.927400386350968, 24.928068483287923, 24.928068483288385, 69.05411764705883]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (2, 2, 2)
  - $q$ : 30.932783
  - $\hat{q}^\beta$ : 7.746030
  - $q^\beta$ : 7.838833
  - Breakpoints: [0, 18.89081402008292, 22.02028770105733, 41.49476194796521, 76.14235294117645]
  - Error: 0.092802594040601
  - Relative error: 0.011838828257388
- (2, 5, 2)
  - $q$ : 14.023794
  - $\hat{q}^\beta$ : 3.564264
  - $q^\beta$ : 4.876624
  - Breakpoints: [0, 2.702328746408619, 4.0106608195978595, 138.54823529411763]
  - Error: 1.312360664799733
  - Relative error: 0.269112514788232
- (3, 1, 1)
  - $q$ : 1.221215
  - $\hat{q}^\beta$ : 0.428186
  - $q^\beta$ : 1.127393
  - Breakpoints: [0, 13.737197536033536, 89.41176470588236]
  - Error: 0.699206366059890
  - Relative error: 0.620197748416862
- (3, 5, 2)
  - $q$ : 41.098456
  - $\hat{q}^\beta$ : 9.166515
  - $q^\beta$ : 9.295983
  - Breakpoints: [0, 8.941176470587976, 27.976182369080462, 55.82566076118323, 71.16597394333384, 103.45829612760171, 181.68117647058827]
  - Error: 0.129467568869332

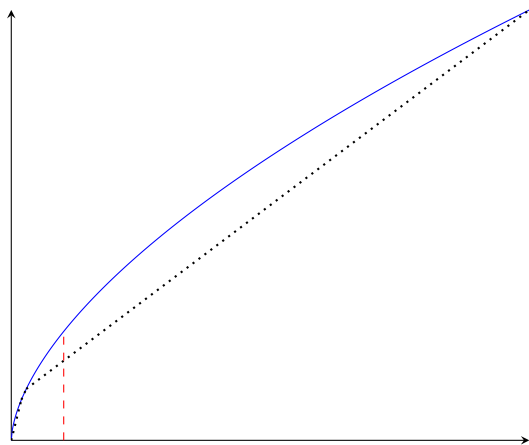
- Relative error: 0.013927260222823
- (4, 3, 2)
  - $q$ : 23.668655
  - $\hat{q}^\beta$ : 6.675779
  - $q^\beta$ : 6.675813
  - Breakpoints: [0, 19.92063386984238, 23.66863193022334, 23.668654734065058, 23.668855770059167, 23.668855770059224, 181.14352941176475]
  - Error: 0.000034118307918
  - Relative error: 0.000005110734730
- (5, 4, 2)
  - $q$ : 82.719688
  - $\hat{q}^\beta$ : 14.143775
  - $q^\beta$ : 14.143775
  - Breakpoints: [0, 11.133625686778567, 51.43476571725509, 82.7196877037192, 82.71968770371944, 106.27900213658211, 192.3352941176471]
  - Error: 0.0000000000000004
  - Relative error: 0.0000000000000000
- $cu, 3$ 
  - $q$ : 20.613988
  - $\hat{q}^\beta$ : 6.026388
  - $q^\beta$ : 6.144651
  - Breakpoints: [0, 7.499856372317694, 10.320104112088831, 13.6435791010403, 13.643579101040302, 14.376341772690381, 16.738084386188994, 16.73892021814917, 16.73899708924428, 16.738997089244634, 16.738997089244645, 18.852085704732907, 128.44338870924773]
  - Error: 0.118262821242016
  - Relative error: 0.019246467110557
- $cu, 5$ 
  - $q$ : 16.234929
  - $\hat{q}^\beta$ : 5.324395
  - $q^\beta$ : 5.324395
  - Breakpoints: [0, 6.717165864481, 9.832103793393394, 14.682070969056184, 16.23492858720698, 16.234975184591587, 16.235135805353295, 20.944873037188774, 20.94487303718885, 20.944873037188852, 21.50633939847323, 22.410514202757604, 127.1036185995295]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000



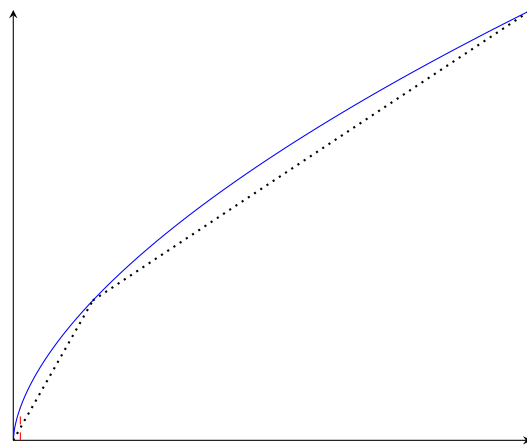
(a)  $q^\beta - (1, 1, 2)$



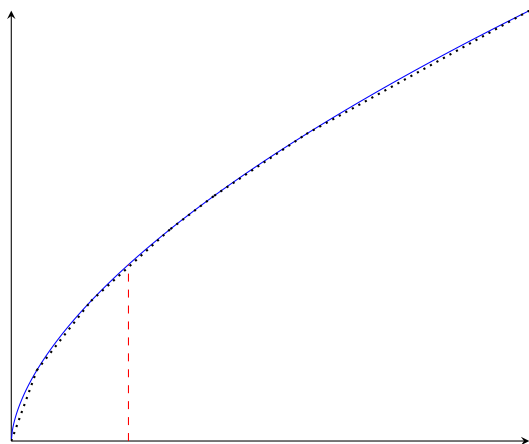
(b)  $q^\beta - (2, 2, 2)$



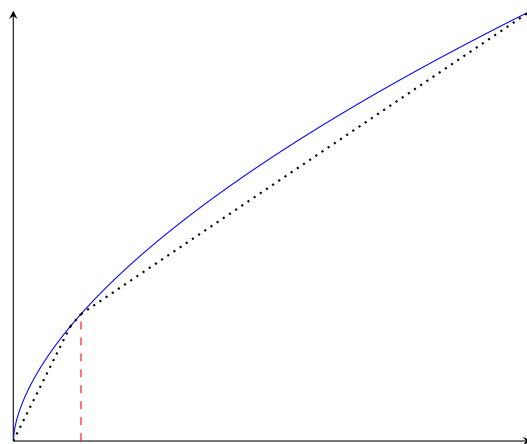
(c)  $q^\beta - (2, 5, 2)$



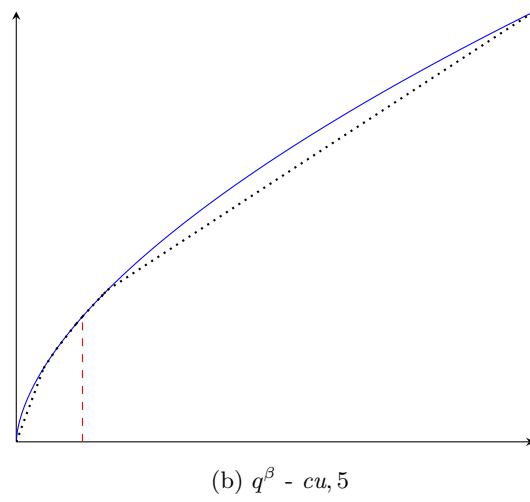
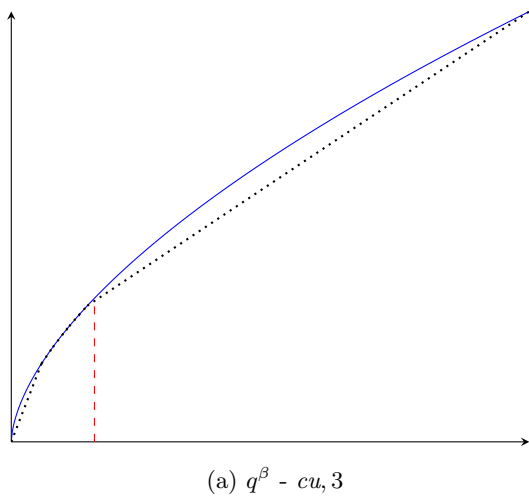
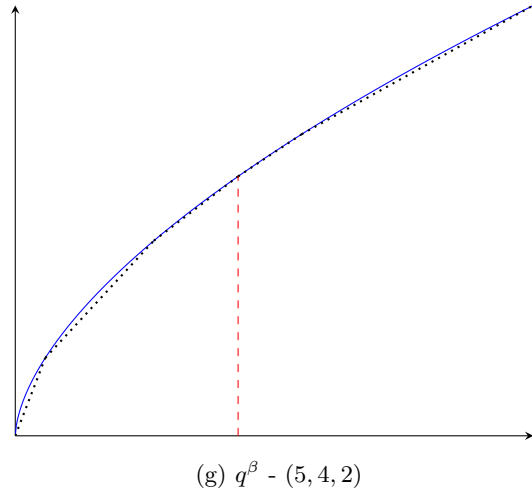
(d)  $q^\beta - (3, 1, 1)$



(e)  $q^\beta - (3, 5, 2)$



(f)  $q^\beta - (4, 3, 2)$



## 7 Other variables

| Variable    | Value | Variable          | Value | Variable          | Value |
|-------------|-------|-------------------|-------|-------------------|-------|
| $z_{1,1,1}$ | -0.0  | $z_{4,5,2}$       | -0.0  | $A_{2,4,2}^\beta$ | 0.0   |
| $z_{1,2,1}$ | 0.0   | $z_{5,1,1}$       | 0.0   | $A_{2,5,1}^\beta$ | 0.0   |
| $z_{1,2,2}$ | -0.0  | $z_{5,1,2}$       | -0.0  | $A_{3,1,2}^\beta$ | 0.0   |
| $z_{1,3,1}$ | 0.0   | $z_{5,2,1}$       | 0.0   | $A_{3,2,1}^\beta$ | 0.0   |
| $z_{1,3,2}$ | -0.0  | $z_{5,2,2}$       | -0.0  | $A_{3,2,2}^\beta$ | 0.0   |
| $z_{1,4,1}$ | 0.0   | $z_{5,3,1}$       | 0.0   | $A_{3,3,1}^\beta$ | 0.0   |
| $z_{1,4,2}$ | -0.0  | $z_{5,3,2}$       | -0.0  | $A_{3,3,2}^\beta$ | 0.0   |
| $z_{1,5,1}$ | 0.0   | $z_{5,4,1}$       | -0.0  | $A_{3,4,1}^\beta$ | 0.0   |
| $z_{1,5,2}$ | -0.0  | $z_{5,5,1}$       | 0.0   | $A_{3,4,2}^\beta$ | 0.0   |
| $z_{2,1,1}$ | -0.0  | $z_{5,5,2}$       | -0.0  | $A_{3,5,1}^\beta$ | 0.0   |
| $z_{2,1,2}$ | -0.0  | $z_{cu,1}$        | -0.0  | $A_{4,1,1}^\beta$ | 0.0   |
| $z_{2,2,1}$ | 0.0   | $z_{cu,2}$        | -0.0  | $A_{4,1,2}^\beta$ | 0.0   |
| $z_{2,3,1}$ | -0.0  | $z_{cu,4}$        | -0.0  | $A_{4,2,1}^\beta$ | 0.0   |
| $z_{2,3,2}$ | -0.0  | $z_{hu,1}$        | -0.0  | $A_{4,2,2}^\beta$ | 0.0   |
| $z_{2,4,1}$ | -0.0  | $z_{hu,2}$        | -0.0  | $A_{4,3,1}^\beta$ | 0.0   |
| $z_{2,4,2}$ | 0.0   | $z_{hu,3}$        | -0.0  | $A_{4,4,1}^\beta$ | 0.0   |
| $z_{2,5,1}$ | -0.0  | $z_{hu,4}$        | -0.0  | $A_{4,4,2}^\beta$ | 0.0   |
| $z_{3,1,2}$ | -0.0  | $z_{hu,5}$        | -0.0  | $A_{4,5,1}^\beta$ | 0.0   |
| $z_{3,2,1}$ | -0.0  | $A_{1,1,1}^\beta$ | 0.0   | $A_{4,5,2}^\beta$ | 0.0   |
| $z_{3,2,2}$ | -0.0  | $A_{1,2,1}^\beta$ | 0.0   | $A_{5,1,1}^\beta$ | 0.0   |
| $z_{3,3,1}$ | -0.0  | $A_{1,2,2}^\beta$ | 0.0   | $A_{5,1,2}^\beta$ | 0.0   |
| $z_{3,3,2}$ | -0.0  | $A_{1,3,1}^\beta$ | 0.0   | $A_{5,2,1}^\beta$ | 0.0   |
| $z_{3,4,1}$ | -0.0  | $A_{1,3,2}^\beta$ | 0.0   | $A_{5,2,2}^\beta$ | 0.0   |
| $z_{3,4,2}$ | -0.0  | $A_{1,4,1}^\beta$ | 0.0   | $A_{5,3,1}^\beta$ | 0.0   |
| $z_{3,5,1}$ | -0.0  | $A_{1,4,2}^\beta$ | 0.0   | $A_{5,3,2}^\beta$ | 0.0   |
| $z_{4,1,1}$ | -0.0  | $A_{1,5,1}^\beta$ | 0.0   | $A_{5,4,1}^\beta$ | 0.0   |
| $z_{4,1,2}$ | -0.0  | $A_{1,5,2}^\beta$ | 0.0   | $A_{5,5,1}^\beta$ | 0.0   |
| $z_{4,2,1}$ | -0.0  | $A_{2,1,1}^\beta$ | 0.0   | $A_{5,5,2}^\beta$ | 0.0   |
| $z_{4,2,2}$ | -0.0  | $A_{2,1,2}^\beta$ | 0.0   | $A_{cu,1}^\beta$  | 0.0   |
| $z_{4,3,1}$ | -0.0  | $A_{2,2,1}^\beta$ | 0.0   | $A_{cu,2}^\beta$  | 0.0   |
| $z_{4,4,1}$ | -0.0  | $A_{2,3,1}^\beta$ | 0.0   | $A_{cu,4}^\beta$  | 0.0   |
| $z_{4,4,2}$ | -0.0  | $A_{2,3,2}^\beta$ | 0.0   | $A_{hu,1}^\beta$  | 0.0   |
| $z_{4,5,1}$ | -0.0  | $A_{2,4,1}^\beta$ | 0.0   | $A_{hu,2}^\beta$  | 0.0   |

| Variable         | Value | Variable     | Value             | Variable     | Value |
|------------------|-------|--------------|-------------------|--------------|-------|
| $A_{hu,3}^\beta$ | 0.0   | $A_{4,4,1}$  | 0.0               | $dt_{1,5,3}$ | 66.7  |
| $A_{hu,4}^\beta$ | 0.0   | $A_{4,4,2}$  | 0.0               | $dt_{2,1,1}$ | 111.1 |
| $A_{hu,5}^\beta$ | 0.0   | $A_{4,5,1}$  | 0.0               | $dt_{2,1,2}$ | 10.0  |
| $A_{1,1,1}$      | 0.0   | $A_{4,5,2}$  | 0.0               | $dt_{2,1,3}$ | 100.0 |
| $A_{1,2,1}$      | 0.0   | $A_{5,1,1}$  | 0.0               | $dt_{2,2,1}$ | 111.1 |
| $A_{1,2,2}$      | 0.0   | $A_{5,1,2}$  | 0.0               | $dt_{2,3,1}$ | 111.1 |
| $A_{1,3,1}$      | 0.0   | $A_{5,2,1}$  | 0.0               | $dt_{2,3,2}$ | 10.0  |
| $A_{1,3,2}$      | 0.0   | $A_{5,2,2}$  | 0.0               | $dt_{2,3,3}$ | 183.3 |
| $A_{1,4,1}$      | 0.0   | $A_{5,3,1}$  | 0.0               | $dt_{2,4,1}$ | 111.1 |
| $A_{1,4,2}$      | 0.0   | $A_{5,3,2}$  | 0.0               | $dt_{2,4,2}$ | 111.1 |
| $A_{1,5,1}$      | 0.0   | $A_{5,4,1}$  | 0.0               | $dt_{2,4,3}$ | 94.5  |
| $A_{1,5,2}$      | 0.0   | $A_{5,5,1}$  | 0.0               | $dt_{2,5,1}$ | 111.1 |
| $A_{2,1,1}$      | 0.0   | $A_{5,5,2}$  | 0.0               | $dt_{3,1,3}$ | 10.0  |
| $A_{2,1,2}$      | 0.0   | $A_{cu,1}$   | 0.0               | $dt_{3,2,1}$ | 111.1 |
| $A_{2,2,1}$      | 0.0   | $A_{cu,2}$   | 0.0               | $dt_{3,2,2}$ | 111.1 |
| $A_{2,3,1}$      | 0.0   | $A_{cu,4}$   | 0.0               | $dt_{3,2,3}$ | 10.0  |
| $A_{2,3,2}$      | 0.0   | $A_{hu,1}$   | 0.0               | $dt_{3,3,1}$ | 161.1 |
| $A_{2,4,1}$      | 0.0   | $A_{hu,2}$   | 0.0               | $dt_{3,3,2}$ | 10.0  |
| $A_{2,4,2}$      | 0.0   | $A_{hu,3}$   | 0.0               | $dt_{3,3,3}$ | 188.9 |
| $A_{2,5,1}$      | 0.0   | $A_{hu,4}$   | 0.0               | $dt_{3,4,1}$ | 144.5 |
| $A_{3,1,2}$      | 0.0   | $A_{hu,5}$   | 0.0               | $dt_{3,4,2}$ | 144.5 |
| $A_{3,2,1}$      | 0.0   | $dt_{1,1,1}$ | 66.7              | $dt_{3,4,3}$ | 10.0  |
| $A_{3,2,2}$      | 0.0   | $dt_{1,2,1}$ | 17.15006133296988 | $dt_{3,5,1}$ | 133.4 |
| $A_{3,3,1}$      | 0.0   | $dt_{1,2,2}$ | 17.15006133296988 | $dt_{4,1,1}$ | 122.2 |
| $A_{3,3,2}$      | 0.0   | $dt_{1,2,3}$ | 17.15006133296988 | $dt_{4,1,2}$ | 10.0  |
| $A_{3,4,1}$      | 0.0   | $dt_{1,3,1}$ | 66.7              | $dt_{4,1,3}$ | 100.0 |
| $A_{3,4,2}$      | 0.0   | $dt_{1,3,2}$ | 10.0              | $dt_{4,2,1}$ | 122.2 |
| $A_{3,5,1}$      | 0.0   | $dt_{1,3,3}$ | 122.2             | $dt_{4,2,2}$ | 10.0  |
| $A_{4,1,1}$      | 0.0   | $dt_{1,4,1}$ | 66.7              | $dt_{4,2,3}$ | 106.1 |
| $A_{4,1,2}$      | 0.0   | $dt_{1,4,2}$ | 66.7              | $dt_{4,3,1}$ | 122.2 |
| $A_{4,2,1}$      | 0.0   | $dt_{1,4,3}$ | 77.8              | $dt_{4,4,1}$ | 122.2 |
| $A_{4,2,2}$      | 0.0   | $dt_{1,5,1}$ | 66.7              | $dt_{4,4,2}$ | 10.0  |
| $A_{4,3,1}$      | 0.0   | $dt_{1,5,2}$ | 66.7              | $dt_{4,4,3}$ | 94.5  |



| Variable      | Value              | Variable      | Value              | Variable      | Value             |
|---------------|--------------------|---------------|--------------------|---------------|-------------------|
| $dt_{4,5,1}$  | 122.2              | $f_{2,1,1}^H$ | 10.6               | $f_{5,5,2}^H$ | 5.435558889722431 |
| $dt_{4,5,2}$  | 10.0               | $f_{2,1,2}^H$ | 0.0                | $f_{1,1,1}^C$ | 0.0               |
| $dt_{4,5,3}$  | 111.1              | $f_{2,2,1}^H$ | 0.0                | $f_{1,2,1}^C$ | 0.0               |
| $dt_{5,1,1}$  | 133.3              | $f_{2,3,1}^H$ | 0.0                | $f_{1,2,2}^C$ | 0.0               |
| $dt_{5,1,2}$  | 138.9              | $f_{2,3,2}^H$ | 0.0                | $f_{1,3,1}^C$ | 0.0               |
| $dt_{5,1,3}$  | 10.0               | $f_{2,4,1}^H$ | 0.0                | $f_{1,3,2}^C$ | 0.0               |
| $dt_{5,2,1}$  | 74.66004781348396  | $f_{2,4,2}^H$ | 0.0                | $f_{1,4,1}^C$ | 0.0               |
| $dt_{5,2,2}$  | 74.66004781348396  | $f_{2,5,1}^H$ | 0.0                | $f_{1,4,2}^C$ | 0.0               |
| $dt_{5,2,3}$  | 74.66004781348391  | $f_{3,1,2}^H$ | 0.0                | $f_{1,5,1}^C$ | 0.0               |
| $dt_{5,3,1}$  | 133.3              | $f_{3,2,1}^H$ | 13.725884543761719 | $f_{1,5,2}^C$ | 0.0               |
| $dt_{5,3,2}$  | 53.173208931823815 | $f_{3,2,2}^H$ | 1.6583620952021967 | $f_{2,1,1}^C$ | 0.0               |
| $dt_{5,3,3}$  | 161.1              | $f_{3,3,1}^H$ | 0.0                | $f_{2,1,2}^C$ | 0.0               |
| $dt_{5,4,1}$  | 116.7              | $f_{3,3,2}^H$ | 5.495598252525539  | $f_{2,2,1}^C$ | 6.1               |
| $dt_{5,5,1}$  | 105.6              | $f_{3,4,1}^H$ | 0.0                | $f_{2,3,1}^C$ | 0.0               |
| $dt_{5,5,2}$  | 105.6              | $f_{3,4,2}^H$ | 0.0                | $f_{2,3,2}^C$ | 0.0               |
| $dt_{5,5,3}$  | 105.6              | $f_{3,5,1}^H$ | 0.0                | $f_{2,4,1}^C$ | 0.0               |
| $dt_{cu,1}$   | 53.3               | $f_{4,1,1}^H$ | 12.6               | $f_{2,4,2}^C$ | 0.0               |
| $dt_{cu,2}$   | 97.8               | $f_{4,1,2}^H$ | 0.0                | $f_{2,5,1}^C$ | 13.9              |
| $dt_{cu,4}$   | 108.9              | $f_{4,2,1}^H$ | 0.0                | $f_{3,1,2}^C$ | 0.0               |
| $dt_{hu,1}$   | 80.0               | $f_{4,2,2}^H$ | 0.0                | $f_{3,2,1}^C$ | 0.0               |
| $dt_{hu,2}$   | 18.3               | $f_{4,3,1}^H$ | 0.0                | $f_{3,2,2}^C$ | 0.0               |
| $dt_{hu,3}$   | 18.9               | $f_{4,4,1}^H$ | 0.0                | $f_{3,3,1}^C$ | 0.0               |
| $dt_{hu,4}$   | 63.3               | $f_{4,4,2}^H$ | 0.0                | $f_{3,3,2}^C$ | 0.0               |
| $dt_{hu,5}$   | 35.6               | $f_{4,5,1}^H$ | 0.0                | $f_{3,4,1}^C$ | 0.0               |
| $f_{1,1,1}^H$ | 8.8                | $f_{4,5,2}^H$ | 0.0                | $f_{3,4,2}^C$ | 0.0               |
| $f_{1,2,1}^H$ | 0.0                | $f_{5,1,1}^H$ | 0.0                | $f_{3,5,1}^C$ | 0.0               |
| $f_{1,2,2}^H$ | 0.0                | $f_{5,1,2}^H$ | 0.0                | $f_{4,1,1}^C$ | 0.0               |
| $f_{1,3,1}^H$ | 0.0                | $f_{5,2,1}^H$ | 0.0                | $f_{4,1,2}^C$ | 1.730400000000138 |
| $f_{1,3,2}^H$ | 0.0                | $f_{5,2,2}^H$ | 0.0                | $f_{4,2,1}^C$ | 0.0               |
| $f_{1,4,1}^H$ | 0.0                | $f_{5,3,1}^H$ | 17.7               | $f_{4,2,2}^C$ | 0.0               |
| $f_{1,4,2}^H$ | 0.0                | $f_{5,3,2}^H$ | 0.0                | $f_{4,3,1}^C$ | 8.4               |
| $f_{1,5,1}^H$ | 0.0                | $f_{5,4,1}^H$ | 0.0                | $f_{4,4,1}^C$ | 17.3              |
| $f_{1,5,2}^H$ | 0.0                | $f_{5,5,1}^H$ | 0.0                | $f_{4,4,2}^C$ | 0.0               |

| Variable                 | Value                | Variable                 | Value                | Variable    | Value |
|--------------------------|----------------------|--------------------------|----------------------|-------------|-------|
| $f_{4,5,1}^C$            | 0.0                  | $\text{RecLMTD}_{3,4,1}$ | 0.1                  | $q_{1,3,2}$ | 0.0   |
| $f_{4,5,2}^C$            | 0.0                  | $\text{RecLMTD}_{3,4,2}$ | 0.1                  | $q_{1,4,1}$ | 0.0   |
| $f_{5,1,1}^C$            | 0.0                  | $\text{RecLMTD}_{3,5,1}$ | 0.1                  | $q_{1,4,2}$ | 0.0   |
| $f_{5,1,2}^C$            | 0.0                  | $\text{RecLMTD}_{4,1,1}$ | 0.020350921041481694 | $q_{1,5,1}$ | 0.0   |
| $f_{5,2,1}^C$            | 0.0                  | $\text{RecLMTD}_{4,1,2}$ | 0.02150212335787724  | $q_{1,5,2}$ | 0.0   |
| $f_{5,2,2}^C$            | 0.0                  | $\text{RecLMTD}_{4,2,1}$ | 0.1                  | $q_{2,1,1}$ | 0.0   |
| $f_{5,3,1}^C$            | 0.0                  | $\text{RecLMTD}_{4,2,2}$ | 0.029297885186065308 | $q_{2,1,2}$ | 0.0   |
| $f_{5,3,2}^C$            | 0.0                  | $\text{RecLMTD}_{4,3,1}$ | 0.017133557887648563 | $q_{2,2,1}$ | 0.0   |
| $f_{5,4,1}^C$            | 0.0                  | $\text{RecLMTD}_{4,4,1}$ | 0.02230651890829312  | $q_{2,3,1}$ | 0.0   |
| $f_{5,5,1}^C$            | 0.0                  | $\text{RecLMTD}_{4,4,2}$ | 0.024030191698075007 | $q_{2,3,2}$ | 0.0   |
| $f_{5,5,2}^C$            | 0.0                  | $\text{RecLMTD}_{4,5,1}$ | 0.1                  | $q_{2,4,1}$ | 0.0   |
| $\text{RecLMTD}_{1,1,1}$ | 0.1                  | $\text{RecLMTD}_{4,5,2}$ | 0.02554670259920889  | $q_{2,4,2}$ | 0.0   |
| $\text{RecLMTD}_{1,2,1}$ | 0.1                  | $\text{RecLMTD}_{5,1,1}$ | 0.007344552824542702 | $q_{2,5,1}$ | 0.0   |
| $\text{RecLMTD}_{1,2,2}$ | 0.1                  | $\text{RecLMTD}_{5,1,2}$ | 0.1                  | $q_{3,1,2}$ | 0.0   |
| $\text{RecLMTD}_{1,3,1}$ | 0.1                  | $\text{RecLMTD}_{5,2,1}$ | 0.1                  | $q_{3,2,1}$ | 0.0   |
| $\text{RecLMTD}_{1,3,2}$ | 0.1                  | $\text{RecLMTD}_{5,2,2}$ | 0.1                  | $q_{3,2,2}$ | 0.0   |
| $\text{RecLMTD}_{1,4,1}$ | 0.1                  | $\text{RecLMTD}_{5,3,1}$ | 0.1                  | $q_{3,3,1}$ | 0.0   |
| $\text{RecLMTD}_{1,4,2}$ | 0.013770395384645695 | $\text{RecLMTD}_{5,3,2}$ | 0.028816453654312128 | $q_{3,3,2}$ | 0.0   |
| $\text{RecLMTD}_{1,5,1}$ | 0.1                  | $\text{RecLMTD}_{5,4,1}$ | 0.1                  | $q_{3,4,1}$ | 0.0   |
| $\text{RecLMTD}_{1,5,2}$ | 0.1                  | $\text{RecLMTD}_{5,5,1}$ | 0.1                  | $q_{3,4,2}$ | 0.0   |
| $\text{RecLMTD}_{2,1,1}$ | 0.1                  | $\text{RecLMTD}_{5,5,2}$ | 0.009469696969696968 | $q_{3,5,1}$ | 0.0   |
| $\text{RecLMTD}_{2,1,2}$ | 0.1                  | $\text{RecLMTD}_{cu,1}$  | 0.016108908592626727 | $q_{4,1,1}$ | 0.0   |
| $\text{RecLMTD}_{2,2,1}$ | 0.1                  | $\text{RecLMTD}_{cu,2}$  | 0.009423141115768875 | $q_{4,1,2}$ | 0.0   |
| $\text{RecLMTD}_{2,3,1}$ | 0.1                  | $\text{RecLMTD}_{cu,4}$  | 0.008535688158765427 | $q_{4,2,1}$ | 0.0   |
| $\text{RecLMTD}_{2,3,2}$ | 0.020926522199679467 | $\text{RecLMTD}_{hu,1}$  | 0.012387051800747543 | $q_{4,2,2}$ | 0.0   |
| $\text{RecLMTD}_{2,4,1}$ | 0.1                  | $\text{RecLMTD}_{hu,2}$  | 0.045618305265862594 | $q_{4,3,1}$ | 0.0   |
| $\text{RecLMTD}_{2,4,2}$ | 0.00946741317027322  | $\text{RecLMTD}_{hu,3}$  | 0.04305097282506776  | $q_{4,4,1}$ | 0.0   |
| $\text{RecLMTD}_{2,5,1}$ | 0.1                  | $\text{RecLMTD}_{hu,4}$  | 0.015434474530129737 | $q_{4,4,2}$ | 0.0   |
| $\text{RecLMTD}_{3,1,2}$ | 0.1                  | $\text{RecLMTD}_{hu,5}$  | 0.024141568940966463 | $q_{4,5,1}$ | 0.0   |
| $\text{RecLMTD}_{3,2,1}$ | 0.1                  | $q_{1,1,1}$              | 0.0                  | $q_{4,5,2}$ | 0.0   |
| $\text{RecLMTD}_{3,2,2}$ | 0.1                  | $q_{1,2,1}$              | 0.0                  | $q_{5,1,1}$ | 0.0   |
| $\text{RecLMTD}_{3,3,1}$ | 0.1                  | $q_{1,2,2}$              | 0.0                  | $q_{5,1,2}$ | 0.0   |
| $\text{RecLMTD}_{3,3,2}$ | 0.026450800778310225 | $q_{1,3,1}$              | 0.0                  | $q_{5,2,1}$ | 0.0   |

| Variable      | Value              | Variable      | Value              | Variable      | Value              |
|---------------|--------------------|---------------|--------------------|---------------|--------------------|
| $q_{5,2,2}$   | 0.0                | $t_{3,2,2}^H$ | 79.3233694336735   | $t_{2,1,1}^C$ | 129.7960526315789  |
| $q_{5,3,1}$   | 0.0                | $t_{3,3,1}^H$ | 226.7              | $t_{2,1,2}^C$ | 150.87721563365653 |
| $q_{5,3,2}$   | 0.0                | $t_{3,3,2}^H$ | 131.03056111564754 | $t_{2,2,1}^C$ | 221.7              |
| $q_{5,4,1}$   | 0.0                | $t_{3,4,1}^H$ | 226.7              | $t_{2,3,1}^C$ | 37.8               |
| $q_{5,5,1}$   | 0.0                | $t_{3,4,2}^H$ | 76.03027775241064  | $t_{2,3,2}^C$ | 37.8               |
| $q_{5,5,2}$   | 0.0                | $t_{3,5,1}^H$ | 72.64443425981113  | $t_{2,4,1}^C$ | 82.2               |
| $q_{cu,1}$    | 0.0                | $t_{4,1,1}^H$ | 271.1              | $t_{2,4,2}^C$ | 82.2               |
| $q_{cu,2}$    | 0.0                | $t_{4,1,2}^H$ | 148.9              | $t_{2,5,1}^C$ | 204.4              |
| $q_{cu,4}$    | 0.0                | $t_{4,2,1}^H$ | 204.06666666666666 | $t_{3,1,2}^C$ | 90.20394736842073  |
| $q_{hu,1}$    | 0.0                | $t_{4,2,2}^H$ | 148.9              | $t_{3,2,1}^C$ | 221.7              |
| $q_{hu,2}$    | 0.0                | $t_{4,3,1}^H$ | 271.1              | $t_{3,2,2}^C$ | 115.6              |
| $q_{hu,3}$    | 0.0                | $t_{4,4,1}^H$ | 271.1              | $t_{3,3,1}^C$ | 111.5              |
| $q_{hu,4}$    | 0.0                | $t_{4,4,2}^H$ | 148.9              | $t_{3,3,2}^C$ | 37.8               |
| $q_{hu,5}$    | 0.0                | $t_{4,5,1}^H$ | 271.1              | $t_{3,4,1}^C$ | 176.7              |
| $t_{1,1,1}^H$ | 160.0              | $t_{4,5,2}^H$ | 148.9              | $t_{3,4,2}^C$ | 82.2               |
| $t_{1,2,1}^H$ | 160.0              | $t_{5,1,1}^H$ | 189.12372881355932 | $t_{3,5,1}^C$ | 204.4              |
| $t_{1,2,2}^H$ | 160.0              | $t_{5,1,2}^H$ | 65.6006971636449   | $t_{4,1,1}^C$ | 60.0               |
| $t_{1,3,1}^H$ | 160.0              | $t_{5,2,1}^H$ | 198.9              | $t_{4,1,2}^C$ | 60.0               |
| $t_{1,3,2}^H$ | 160.0              | $t_{5,2,2}^H$ | 65.60093358726036  | $t_{4,2,1}^C$ | 221.7              |
| $t_{1,4,1}^H$ | 160.0              | $t_{5,3,1}^H$ | 198.9              | $t_{4,2,2}^C$ | 115.6              |
| $t_{1,4,2}^H$ | 160.0              | $t_{5,3,2}^H$ | 65.60050795803234  | $t_{4,3,1}^C$ | 221.1              |
| $t_{1,5,1}^H$ | 160.0              | $t_{5,4,1}^H$ | 198.9              | $t_{4,4,1}^C$ | 176.7              |
| $t_{1,5,2}^H$ | 160.0              | $t_{5,5,1}^H$ | 198.9              | $t_{4,4,2}^C$ | 82.2               |
| $t_{2,1,1}^H$ | 248.9              | $t_{5,5,2}^H$ | 198.9              | $t_{4,5,1}^C$ | 204.4              |
| $t_{2,1,2}^H$ | 169.0717863049703  | $t_{1,1,1}^C$ | 90.20394736842101  | $t_{4,5,2}^C$ | 93.3               |
| $t_{2,2,1}^H$ | 139.85985526856254 | $t_{1,2,1}^C$ | 221.7              | $t_{5,1,1}^C$ | 92.62135922330097  |
| $t_{2,3,1}^H$ | 248.9              | $t_{1,2,2}^C$ | 115.6              | $t_{5,1,2}^C$ | 90.20394736842104  |
| $t_{2,3,2}^H$ | 180.14061952507436 | $t_{1,3,1}^C$ | 111.5              | $t_{5,2,1}^C$ | 221.7              |
| $t_{2,4,1}^H$ | 198.8575471698113  | $t_{1,3,2}^C$ | 107.67619047619047 | $t_{5,2,2}^C$ | 115.6              |
| $t_{2,4,2}^H$ | 197.05218912643699 | $t_{1,4,1}^C$ | 176.7              | $t_{5,3,1}^C$ | 111.5              |
| $t_{2,5,1}^H$ | 248.9              | $t_{1,4,2}^C$ | 129.3704699676358  | $t_{5,3,2}^C$ | 37.8               |
| $t_{3,1,2}^H$ | 65.6               | $t_{1,5,1}^C$ | 204.4              | $t_{5,4,1}^C$ | 176.7              |
| $t_{3,2,1}^H$ | 226.7              | $t_{1,5,2}^C$ | 93.3               | $t_{5,5,1}^C$ | 204.4              |

| Variable           | Value              | Variable            | Value              | Variable            | Value              |
|--------------------|--------------------|---------------------|--------------------|---------------------|--------------------|
| $t_{5,5,2}^C$      | 124.01079136690645 | $b_{4,5,1}^{H,in}$  | 0.0                | $b_{3,4,1}^{H,out}$ | 0.0                |
| $b_{1,1,1}^{H,in}$ | 1408.0             | $b_{4,5,2}^{H,in}$  | 0.0                | $b_{3,4,2}^{H,out}$ | 0.0                |
| $b_{1,2,1}^{H,in}$ | 0.0                | $b_{5,1,1}^{H,in}$  | 0.0                | $b_{3,5,1}^{H,out}$ | 0.0                |
| $b_{1,2,2}^{H,in}$ | 0.0                | $b_{5,1,2}^{H,in}$  | 0.0                | $b_{4,1,1}^{H,out}$ | 3415.86            |
| $b_{1,3,1}^{H,in}$ | 0.0                | $b_{5,2,1}^{H,in}$  | 0.0                | $b_{4,1,2}^{H,out}$ | 0.0                |
| $b_{1,3,2}^{H,in}$ | 0.0                | $b_{5,2,2}^{H,in}$  | 0.0                | $b_{4,2,1}^{H,out}$ | 0.0                |
| $b_{1,4,1}^{H,in}$ | 0.0                | $b_{5,3,1}^{H,in}$  | 3520.5300000000007 | $b_{4,2,2}^{H,out}$ | 0.0                |
| $b_{1,4,2}^{H,in}$ | 0.0                | $b_{5,3,2}^{H,in}$  | 0.0                | $b_{4,3,1}^{H,out}$ | 0.0                |
| $b_{1,5,1}^{H,in}$ | 0.0                | $b_{5,4,1}^{H,in}$  | 0.0                | $b_{4,4,1}^{H,out}$ | 0.0                |
| $b_{1,5,2}^{H,in}$ | 0.0                | $b_{5,5,1}^{H,in}$  | 0.0                | $b_{4,4,2}^{H,out}$ | 0.0                |
| $b_{2,1,1}^{H,in}$ | 2638.34            | $b_{5,5,2}^{H,in}$  | 1081.1326631657917 | $b_{4,5,1}^{H,out}$ | 0.0                |
| $b_{2,1,2}^{H,in}$ | 0.0                | $b_{1,1,1}^{H,out}$ | 1408.0             | $b_{4,5,2}^{H,out}$ | 0.0                |
| $b_{2,2,1}^{H,in}$ | 0.0                | $b_{1,2,1}^{H,out}$ | 0.0                | $b_{5,1,1}^{H,out}$ | 0.0                |
| $b_{2,3,1}^{H,in}$ | 0.0                | $b_{1,2,2}^{H,out}$ | 0.0                | $b_{5,1,2}^{H,out}$ | 0.0                |
| $b_{2,3,2}^{H,in}$ | 0.0                | $b_{1,3,1}^{H,out}$ | 0.0                | $b_{5,2,1}^{H,out}$ | 0.0                |
| $b_{2,4,1}^{H,in}$ | 0.0                | $b_{1,3,2}^{H,out}$ | 0.0                | $b_{5,2,2}^{H,out}$ | 0.0                |
| $b_{2,4,2}^{H,in}$ | 0.0                | $b_{1,4,1}^{H,out}$ | 0.0                | $b_{5,3,1}^{H,out}$ | 3520.5300000000007 |
| $b_{2,5,1}^{H,in}$ | 0.0                | $b_{1,4,2}^{H,out}$ | 0.0                | $b_{5,3,2}^{H,out}$ | 0.0                |
| $b_{3,1,2}^{H,in}$ | 0.0                | $b_{1,5,1}^{H,out}$ | 0.0                | $b_{5,4,1}^{H,out}$ | 0.0                |
| $b_{3,2,1}^{H,in}$ | 3111.6580260707815 | $b_{1,5,2}^{H,out}$ | 0.0                | $b_{5,5,1}^{H,out}$ | 0.0                |
| $b_{3,2,2}^{H,in}$ | 202.91068698235074 | $b_{2,1,1}^{H,out}$ | 2638.34            | $b_{5,5,2}^{H,out}$ | 1081.1326631657917 |
| $b_{3,3,1}^{H,in}$ | 0.0                | $b_{2,1,2}^{H,out}$ | 0.0                | $b_{1,1,1}^{C,in}$  | 0.0                |
| $b_{3,3,2}^{H,in}$ | 1245.8521238475398 | $b_{2,2,1}^{H,out}$ | 0.0                | $b_{1,2,1}^{C,in}$  | 0.0                |
| $b_{3,4,1}^{H,in}$ | 0.0                | $b_{2,3,1}^{H,out}$ | 0.0                | $b_{1,2,2}^{C,in}$  | 0.0                |
| $b_{3,4,2}^{H,in}$ | 0.0                | $b_{2,3,2}^{H,out}$ | 0.0                | $b_{1,3,1}^{C,in}$  | 0.0                |
| $b_{3,5,1}^{H,in}$ | 0.0                | $b_{2,4,1}^{H,out}$ | 0.0                | $b_{1,3,2}^{C,in}$  | 0.0                |
| $b_{4,1,1}^{H,in}$ | 3415.86            | $b_{2,4,2}^{H,out}$ | 0.0                | $b_{1,4,1}^{C,in}$  | 0.0                |
| $b_{4,1,2}^{H,in}$ | 0.0                | $b_{2,5,1}^{H,out}$ | 0.0                | $b_{1,4,2}^{C,in}$  | 0.0                |
| $b_{4,2,1}^{H,in}$ | 0.0                | $b_{3,1,2}^{H,out}$ | 0.0                | $b_{1,5,1}^{C,in}$  | 0.0                |
| $b_{4,2,2}^{H,in}$ | 0.0                | $b_{3,2,1}^{H,out}$ | 3111.6580260707815 | $b_{1,5,2}^{C,in}$  | 0.0                |
| $b_{4,3,1}^{H,in}$ | 0.0                | $b_{3,2,2}^{H,out}$ | 202.91068698235074 | $b_{2,1,1}^{C,in}$  | 0.0                |
| $b_{4,4,1}^{H,in}$ | 0.0                | $b_{3,3,1}^{H,out}$ | 0.0                | $b_{2,1,2}^{C,in}$  | 0.0                |
| $b_{4,4,2}^{H,in}$ | 0.0                | $b_{3,3,2}^{H,out}$ | 1245.8521238475398 | $b_{2,2,1}^{C,in}$  | 1352.37            |

| Variable            | Value              | Variable            | Value              | Variable            | Value |
|---------------------|--------------------|---------------------|--------------------|---------------------|-------|
| $b_{2,3,1}^{C,in}$  | 0.0                | $b_{1,2,2}^{C,out}$ | 0.0                | $b_{5,1,2}^{C,out}$ | 0.0   |
| $b_{2,3,2}^{C,in}$  | 0.0                | $b_{1,3,1}^{C,out}$ | 0.0                | $b_{5,2,1}^{C,out}$ | 0.0   |
| $b_{2,4,1}^{C,in}$  | 0.0                | $b_{1,3,2}^{C,out}$ | 0.0                | $b_{5,2,2}^{C,out}$ | 0.0   |
| $b_{2,4,2}^{C,in}$  | 0.0                | $b_{1,4,1}^{C,out}$ | 0.0                | $b_{5,3,1}^{C,out}$ | 0.0   |
| $b_{2,5,1}^{C,in}$  | 2841.1600000000003 | $b_{1,4,2}^{C,out}$ | 0.0                | $b_{5,3,2}^{C,out}$ | 0.0   |
| $b_{3,1,2}^{C,in}$  | 0.0                | $b_{1,5,1}^{C,out}$ | 0.0                | $b_{5,4,1}^{C,out}$ | 0.0   |
| $b_{3,2,1}^{C,in}$  | 0.0                | $b_{1,5,2}^{C,out}$ | 0.0                | $b_{5,5,1}^{C,out}$ | 0.0   |
| $b_{3,2,2}^{C,in}$  | 0.0                | $b_{2,1,1}^{C,out}$ | 0.0                | $b_{5,5,2}^{C,out}$ | 0.0   |
| $b_{3,3,1}^{C,in}$  | 0.0                | $b_{2,1,2}^{C,out}$ | 0.0                |                     |       |
| $b_{3,3,2}^{C,in}$  | 0.0                | $b_{2,2,1}^{C,out}$ | 1352.37            |                     |       |
| $b_{3,4,1}^{C,in}$  | 0.0                | $b_{2,3,1}^{C,out}$ | 0.0                |                     |       |
| $b_{3,4,2}^{C,in}$  | 0.0                | $b_{2,3,2}^{C,out}$ | 0.0                |                     |       |
| $b_{3,5,1}^{C,in}$  | 0.0                | $b_{2,4,1}^{C,out}$ | 0.0                |                     |       |
| $b_{4,1,1}^{C,in}$  | 0.0                | $b_{2,4,2}^{C,out}$ | 0.0                |                     |       |
| $b_{4,1,2}^{C,in}$  | 103.82400000000828 | $b_{2,5,1}^{C,out}$ | 2841.1600000000003 |                     |       |
| $b_{4,2,1}^{C,in}$  | 0.0                | $b_{3,1,2}^{C,out}$ | 0.0                |                     |       |
| $b_{4,2,2}^{C,in}$  | 0.0                | $b_{3,2,1}^{C,out}$ | 0.0                |                     |       |
| $b_{4,3,1}^{C,in}$  | 1857.24            | $b_{3,2,2}^{C,out}$ | 0.0                |                     |       |
| $b_{4,4,1}^{C,in}$  | 3056.91            | $b_{3,3,1}^{C,out}$ | 0.0                |                     |       |
| $b_{4,4,2}^{C,in}$  | 0.0                | $b_{3,3,2}^{C,out}$ | 0.0                |                     |       |
| $b_{4,5,1}^{C,in}$  | 0.0                | $b_{3,4,1}^{C,out}$ | 0.0                |                     |       |
| $b_{4,5,2}^{C,in}$  | 0.0                | $b_{3,4,2}^{C,out}$ | 0.0                |                     |       |
| $b_{5,1,1}^{C,in}$  | 0.0                | $b_{3,5,1}^{C,out}$ | 0.0                |                     |       |
| $b_{5,1,2}^{C,in}$  | 0.0                | $b_{4,1,1}^{C,out}$ | 0.0                |                     |       |
| $b_{5,2,1}^{C,in}$  | 0.0                | $b_{4,1,2}^{C,out}$ | 103.82400000000828 |                     |       |
| $b_{5,2,2}^{C,in}$  | 0.0                | $b_{4,2,1}^{C,out}$ | 0.0                |                     |       |
| $b_{5,3,1}^{C,in}$  | 0.0                | $b_{4,2,2}^{C,out}$ | 0.0                |                     |       |
| $b_{5,3,2}^{C,in}$  | 0.0                | $b_{4,3,1}^{C,out}$ | 1857.24            |                     |       |
| $b_{5,4,1}^{C,in}$  | 0.0                | $b_{4,4,1}^{C,out}$ | 3056.91            |                     |       |
| $b_{5,5,1}^{C,in}$  | 0.0                | $b_{4,4,2}^{C,out}$ | 0.0                |                     |       |
| $b_{5,5,2}^{C,in}$  | 0.0                | $b_{4,5,1}^{C,out}$ | 0.0                |                     |       |
| $b_{1,1,1}^{C,out}$ | 0.0                | $b_{4,5,2}^{C,out}$ | 0.0                |                     |       |
| $b_{1,2,1}^{C,out}$ | 0.0                | $b_{5,1,1}^{C,out}$ | 0.0                |                     |       |